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Towards Persistence: Learning Topological Constraints for Event-based Small Object Detection

Shiman He*, Nuo Chen*, Xinyi Ying, Yihang Luo, Yangsi Shi, Zaiping Lin, Miao Li†
National University of Defense Technology
{heshiman24, chennuo97, lm8866}@nudt.edu.cn

Abstract

Small object detection (SOD) plays a vital role in applications such as anti-UAV tasks, yet conventional image-based methods struggle in high-speed scenarios due to the limited frame rate. Event cameras offer a promising alternative by capturing spatiotemporal event streams with microsecond-level temporal resolution. To address the inherent sparsity of small objects in event data, existing methods typically formulate the detection task as semantic segmentation on spatiotemporal point clouds to leverage long-term contextual information. However, these methods often fail to enforce effective spatiotemporal consistency constraints, resulting in fragmented object trajectories. To mitigate these problems, we propose a topology-constrained sparse convolutional network (SpTopoNet), which models the topological structure of moving object trajectories in event point clouds. Our network comprises two key components: a Topological Learning Module (TLM) that discriminates local structures to separate genuine targets from noise, and a Spatial Consistency Module (SCM) that captures long-range spatiotemporal dependencies to enhance trajectory continuity. Additionally, we introduce an event topology-aware loss function that leverages topological correlations to guide the network to maintain structural integrity of target event patterns. Experiments on the benchmark dataset demonstrate the superiority of our method in both detection performance and trajectory completeness.

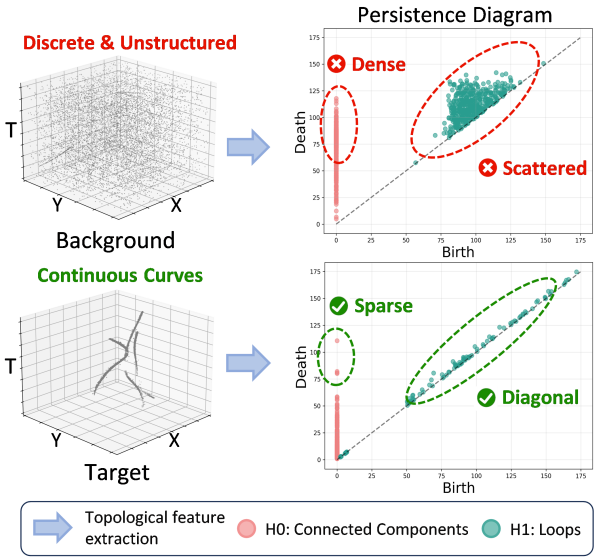


Figure 1. Topological difference between continuous trajectories and discrete backgrounds in real event data. We employ persistent homology analysis. Specifically, we construct gradually expanding spheres centered at each event. Initially ($r = 0$), each event is an independent connected component (H_0 feature). As the radius increases and two spheres intersect, the corresponding events connect and an H_0 feature dies; simultaneously, connecting edges may form closed loops and H_1 features are born. Eventually, when the radius is sufficiently large, all loops are filled and H_1 features die. We record these birth-death processes in a persistence diagram, where the x and y axes represent the birth and death radii of features.

1. Introduction

Small object detection (SOD) is a critical research direction in computer vision that addresses the recognition and localization of small-scale objects in complex scenes [15, 16, 18–21, 24, 51–53], with applications in domains such as anti-UAV systems. However, small moving targets typically lack distinctive appearance features and exhibit complex non-linear motion trajectories, making robust detection in cluttered backgrounds a significant challenge.

Traditional frame-based cameras, constrained by limited frame rates (typically 30 Hz to 60 Hz) and dynamic range (≈60 dB), suffer from motion blur and temporal information loss when capturing high-speed small targets. This significantly degrades detection performance in critical applications. Event cameras [3, 5, 34, 40], functioning as bio-inspired sensors, provide an alternative approach by asynchronously capturing pixel-level brightness changes

* The first two authors contributed equally. † Corresponding author.

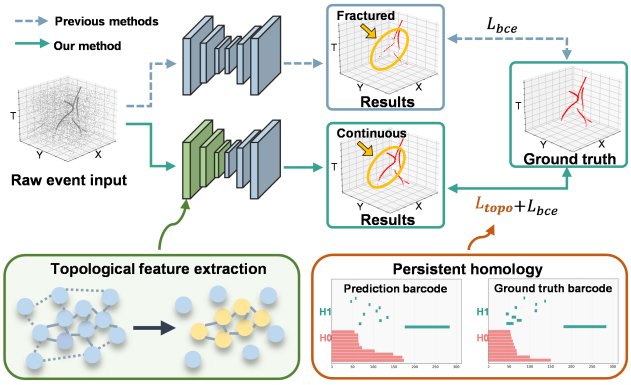


Figure 2. Comparison between our method and previous methods. Our method incorporates topological modeling in both network architecture and loss function, resulting in detection trajectories with greater continuity and completeness compared to previous methods.

as sparse event streams. This unique mechanism grants them high temporal resolution (up to 10^6 Hz), high dynamic range (up to 120 dB), and low data redundancy [8], making them well-suited for dynamic scene perception.

Despite these advantages, event-based SOD presents the following challenges: (1) Small targets: The small size of targets results in insufficient appearance cues, hindering effective feature learning. (2) Complex motion: Objects often follow diverse and non-linear trajectories, making them difficult to track and detect accurately. (3) Asynchronous event streams: The asynchronous nature of event streams further complicates the modeling of these complex trajectories. Among existing methods, frame-based approaches [28, 54] accumulate events into synchronous frames at the cost of destroying the native continuity and sparsity of event data, which results in the loss of temporal precision, motion blur, and redundant processing of static backgrounds. Voxel-based approaches [2, 37, 55] discretize events into spatiotemporal grids, inevitably compromising temporal accuracy and losing fine-grained structural information through quantization. Point cloud-based representations [1, 4, 12, 44] preserve complete asynchronous information in precise (x, y, t, p) coordinates, maintaining native temporal resolution. However, they focus on local spatiotemporal feature aggregation, capturing short-range patterns while neglecting global trajectory structure. This leads to discontinuous detections that fail to maintain long-term coherence across target motion sequences.

The motivation of our work is that the motion trajectories of small targets exhibit specific topological structures in event point clouds that are distinguishable from noise. Events generated by the same target maintain consistent spatiotemporal proximity, forming connected manifolds in (x, y, t) coordinates. As shown in Fig. 1, the target trajectory, appearing as a continuous curve in the event point clouds, exhibits sparse persistent H_0 features and negligible H_1 features. In contrast, discrete and unstructured background noise corresponds to dense H_0 features and spurious persistent H_1 features. Explicitly modeling the topological features can distinguish genuine target trajectories from background noise.

Based on this, we propose SpTopoNet, which maintains trajectory continuity by incorporating topology-aware constraints in both its network architecture and the loss function, as shown in Fig. 2. On one hand, our network captures local spatiotemporal patterns within trajectory segments using the Topological Learning Module (TLM) and models long-range dependencies in the event point clouds via the Spatial Consistency Module (SCM). On the other hand, the EvTopoLoss function regularizes the learning process by penalizing structural discontinuities to enforce global topological consistency of the trajectories. We conduct a comprehensive comparison with state-of-the-art methods on the EVSOD benchmark dataset, demonstrating the superiority of our method. The main contributions are summarized as follows:

- We propose a topology-constrained framework for event-based SOD that models local trajectory topology through explicit spatiotemporal structure learning, achieving state-of-the-art detection performance.
- We introduce EvTopoLoss, a novel optimization objective that maintains global trajectory topology by constraining structural relationships in event point clouds.
- Extensive evaluation on the EV-UAV benchmark [4] demonstrates that our method achieves 11.44% and 5.83% improvements in IoU and detection rate, respectively, over state-of-the-art methods.

2. Related Work

2.1. Object Detection with Event Cameras

Existing event-based object detection methods are primarily categorized into dense and sparse approaches based on event representation. The first category transforms event streams into **dense representations** to leverage conventional convolutional networks. Early methods accumulated events into frame-based representations [28, 54], directly applying image-based detection methods to these event frames. To better preserve temporal information, subsequent methods introduced voxel grid representations [2, 37, 55], which discretize events into multi-channel tensors by quantizing the temporal dimension. These early dense methods typically considered events only within short time windows, resulting in limited temporal context and constrained performance. To capture longer-term temporal dependencies, subsequent works [17, 33] incorporated recurrent neural networks (RNN), which improved detection by

modeling temporal dynamics across consecutive frames. Recent Transformer-based methods [9,10,32,50] further enhanced long-range modeling through self-attention mechanisms. Despite these advancements, dense representation approaches inevitably sacrifice the native microsecond-level temporal precision and introduce motion blur during the densification process. Moreover, converting sparse events into dense tensors introduces redundant background data and consumes additional computational resources.

The second category processes event streams using **sparse representations** to preserve the high temporal resolution and achieve computational efficiency. Graph-based methods [22, 27, 39] model events as dynamic spatiotemporal graphs and aggregate local features through message passing within sparse neighborhood connections. Spiking Neural Networks (SNN) [7, 29, 30, 41] process asynchronous event spikes directly through neuromorphic architectures, transmitting only discrete spike signals between neurons for reduced power consumption. However, these methods often rely on specialized hardware or suffer from suboptimal performance in complex scenarios. More recently, point cloud-based methods [1, 4, 12, 44] have explored modeling events as 3D spatiotemporal point clouds, utilizing sparse convolutions to avoid densification overhead. While these methods successfully preserve the sparse nature of events, they primarily focus on local neighborhood aggregation and lack explicit modeling of global trajectory continuity. Consequently, detection results suffer from trajectory discontinuity, with frequent missed detections at turning points and under occlusions.

To address this issue, we propose SpTopoNet, which leverages topological constraints within a sparse framework to directly segment trajectories from event point clouds. Our method avoids the computational cost associated with dense representations and strengthens topological constraints within sparse representations, leading to improved topological consistency and detection performance.

2.2. Topological Segmentation Loss

Current topological segmentation loss functions typically use topological features based on persistent homology to constrain image segmentation results. The pioneering topology-preserving loss [13] introduced persistent homology into deep learning, enforcing topological consistency between predicted segmentations and ground truth through persistence diagram matching. Building upon this foundation, TopoSeg [11] filters and matches the longest persistent barcodes to handle densely touching and overlapping instances. To address matching ambiguities inherent in persistence diagram matching, Betti matching [42] employs induced matching to construct a common ambient space, enabling both topologically and feature-wise accurate matching. More recently, spatial-aware topological loss [48] leverages information from the image's original spatial domain to assist persistent feature matching, addressing the ambiguous matching problem that arises when relying solely on topological space information.

Among the explorations in point clouds, PHGCN [49] proposes a Persistence Diagram Loss that integrates persistent homology into graph convolution networks for fine-grained 3D object segmentation, demonstrating the feasibility of using persistent homology-based loss functions in point clouds. DSCNet [36] introduces the use of Hausdorff distance to identify and constrain anomalous topological features, demonstrating effectiveness on both 2D and 3D tubular structure segmentation tasks. However, applying these topological losses to unstructured point clouds, particularly event point clouds, generally presents the following challenges: firstly, traditional topological matching primarily relies on the birth-death times of features, ignoring their geometric location information in space, which can lead to severe matching ambiguities in event point clouds with inherent spatial coordinates; secondly, the high computational complexity of persistent homology itself is a bottleneck for practical application.

To address these, we introduce a topological loss with geometry-aware gradient computation based on spatial distances to resolve matching ambiguities. An active region mechanism is employed to compute gradients only near threshold boundaries, enabling efficient batch processing of large-scale scenes.

3. Method

Following the established definition of event segmentation [4], the task is to segment events directly in the point cloud space, event by event. Events can be categorized into three types: target events (E_T), background events (E_B), and noise (E_N):

$$E = E_T \cup E_B \cup E_N. \tag{1}$$

E_T represents events from moving objects with spatiotemporally continuous trajectories, E_B denotes events from static backgrounds or camera motion, and E_N comprises random noise scattered across space and time. To extract E_T from the raw event stream, we propose SpTopoNet with a topological constraint loss to explicitly model the overall topological structure of event trajectories, guiding the network to preserve target events with continuous curved trajectory patterns.

3.1. SpTopoNet

Fig.3 (a) illustrates the overall architecture of SpTopoNet, which follows an encoder-decoder paradigm designed to capture multi-scale topological structures in event point clouds. Since direct computation of persistent homology

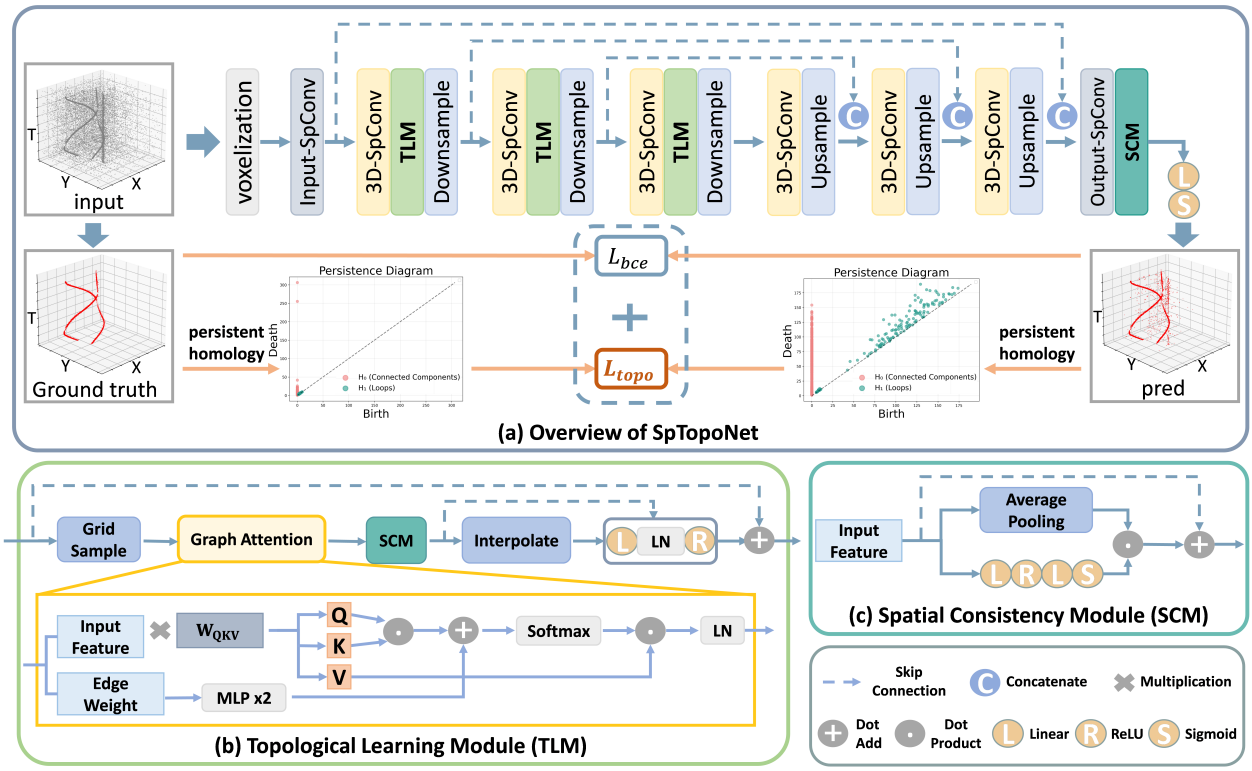


Figure 3. (a) The overview of SpTopoNet. It involves the incorporation of topological constraints into event-based sparse convolutional networks. (b) The Topological Learning Module (TLM) leverages a graph attention mechanism that augments point-wise relationships with edge weight coding to explicitly capture the topological structure. (c) The Spatial Consistency Module (SCM) enhances global feature constraints of the event point clouds.

is costly, we only employ it in the loss function rather than within the network architecture. Instead, we translate topological insights into efficient graph and spatial attention mechanisms, enabling the model to encode such topological principles: Topological Learning Modules (TLM) preserves connectivity via edge-weighted attention, while Spatial Consistency Module (SCM) enforces long-term trajectory coherence, supporting real-time network inference.

The Topological Learning Module. To capture the structural patterns of motion trajectories, we propose TLM to extract topological features from non-uniform event point clouds at local scales, as illustrated in Fig. 3 (b). We adopt an attention-based architecture [46] to encode local topological relationships. Input features are transformed through linear projections to obtain queries $\mathbf{Q}$, keys $\mathbf{K}$, and values $\mathbf{V}$. Additionally, we modulate the attention weights with spatial edge weights:

$$\beta_{ij} = \frac{\exp\left(\frac{\mathbf{Q}_i \mathbf{K}_j^T}{\sqrt{d_k}} + \mathrm{MLP}_{\mathrm{edge}}(w_{ij})\right)}{\sum_{k\in\mathcal{N}(i)} \exp\left(\frac{\mathbf{Q}_i \mathbf{K}_k^T}{\sqrt{d_k}} + \mathrm{MLP}_{\mathrm{edge}}(w_{ik})\right)}, \quad (2)$$

where $w_{ij} = 1/\|\mathbf{x}_i - \mathbf{x}_j\|_2$ encodes inverse distance between neighboring points, $\mathcal{N}(i)$ denotes the k-nearest neighbors of point i. We aggregate features via the modulated attention and apply layer normalization with a residual connection. The output features are upsampled to the original resolution via bilinear interpolation.

The Spatial Consistency Module. To impose long-term topological constraints on trajectory formation, we design SCM (Fig. 3 (c)), which processes input features through two parallel branches. The upper branch extracts global context via average pooling: $\mathbf{F}_{\mathrm{global}} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{F}_{in}^i$, while the lower branch generates spatial attention weights through nonlinear transformations. The output features are obtained by combining these branches as:

$$\mathbf{F}_{\mathrm{out}}^i = \mathbf{F}_{in}^i + \lambda \cdot \left[\sigma(\mathbf{W}_2 \cdot \mathrm{ReLU}(\mathbf{W}_1 \mathbf{F}_{in}^i)) \odot \mathbf{F}_{\mathrm{global}}\right], \quad (3)$$

where $\mathbf{W}_1$ and $\mathbf{W}_2$ are learnable weights, $\sigma(\cdot)$ is the sigmoid function, $\odot$ denotes element-wise multiplication, and λ is a scaling factor. Conditioning global context aggregation on local features enables detail preservation and outlier suppression via residual connections, improving spatiotemporal consistency.

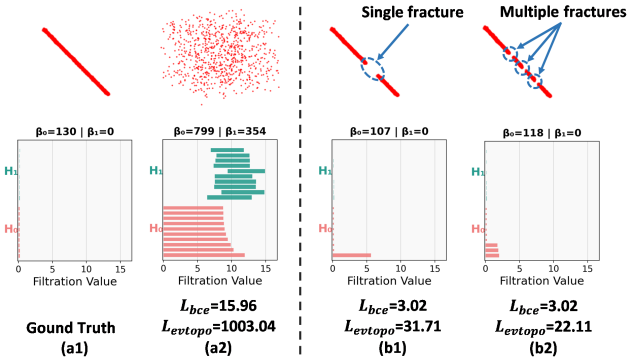


Figure 4. Persistence barcodes for simulated trajectories with equal point counts but distinct topological structures. Each comparison pair (a1-a2, b1-b2) contains equal point counts. (a1) Continuous ground truth, (a2) random noise. (b1) Single fracture, (b2) multiple fractures. $\mathcal{L}_{\text{evtopo}}$ effectively differentiates b1 from b2, while $\mathcal{L}_{\text{bce}}$ remains constant despite the clear topological distinctions.

3.2. EvTopoLoss

3.2.1 Motivation

In general segmentation tasks, the loss function used to train the network is typically the standard pixel-wise binary cross-entropy (BCE) loss:

$$\mathcal{L}_{bce}(p, y) = -[y \cdot \log(p) + (1 - y) \cdot \log(1 - p)]. \quad (4)$$

However, BCE loss focuses solely on pixel-wise classification accuracy, neglecting the global geometric and topological structures of point clouds, such as connectivity, holes, and clustering characteristics, which are essential for accurately characterizing objects in point clouds. As illustrated in Fig. 4, continuous trajectories (a1) and random noise (a2) exhibit fundamentally different structures, as do single-fracture (b1) and multi-fracture (b2) trajectories. However, $\mathcal{L}_{\text{bce}}$ assigns identical values to b1 and b2 despite their structural distinctions. This limitation stems from the pixel-wise formulation's inability to distinguish qualitatively distinct failure modes that compromise trajectory continuity. Specifically, fragmentation, spurious cycles, and discontinuities cannot be adequately penalized without topology-aware loss functions.

3.2.2 Formulation

We introduce a topology-aware loss based on persistent homology to explicitly regularize the network to preserve correct topological structures. Our method utilizes event point clouds derived from predictions and ground truth:

$$P_{\text{pred}} = \left\{e^i | p^i > \tau_{\text{pred}}\right\}, \quad P_{\text{true}} = \left\{e^j | y^j > \tau_{\text{true}}\right\}, \quad (5)$$

where $p \in (0, 1)$ and $y \in \{0, 1\}$ denote predicted probabilities and ground truth labels respectively. The loss comprises two complementary components:

$$\mathcal{L}_{\text{evtopo}} = \mathcal{L}^{(d)}_{\text{betti}} + \mathcal{L}^{(d)}_{\text{wasserstein}}, \quad d = (0, 1), \quad (6)$$

where d indicates the homology dimension (0 for connected components, 1 for loops).

Betti Number Loss enforces correct counts of topological features in each dimension:

$$\mathcal{L}^{(d)}_{\text{betti}} = \begin{cases} 0.5|\Delta\beta_d|^2, & |\Delta\beta_d| \leq 1 \\ |\Delta\beta_d| - 0.5, & |\Delta\beta_d| > 1, \end{cases} \quad (7)$$

where $\Delta\beta_d = \beta_d(P_{\text{pred}}) - \beta_d(P_{\text{true}})$, and $\beta_d(P)$ is the d-dimensional Betti number obtained by counting topological features with persistence exceeding threshold ε. This regularization suppresses spurious fragments and loops introduced by noise (high β_0 and β_1 in Fig. 4 (a2)) while penalizing missing trajectory segments caused by incomplete detection (low β_0 in Fig. 4 (b1-b2)).

Wasserstein Distance Loss ensures not only correct feature counts but also matching significance distributions by comparing persistence diagrams:

$$\mathcal{L}^{(d)}_{\text{wasserstein}} = \sqrt{\frac{1}{K}\sum_{i=1}^{K} w_i \cdot \left|\text{pers}_1^{(i)} - \text{pers}_2^{(i)}\right|^2}, \quad (8)$$

where $pers = death - birth$ quantifies each feature's structural importance. The loss retains the top-K most persistent features and applies exponential decay weights w_i to prioritize prominent structures. While Betti loss ensures the correct number of components, Wasserstein loss guarantees that these components have the correct persistence distribution—preventing scenarios where predictions have the right count but wrong structural prominence.

To balance point-wise accuracy with topological consistency, we combine the topology-aware loss with BCE loss using a weighting factor λ_{topo}:

$$\mathcal{L}_{\text{total}}(p, y) = \mathcal{L}_{\text{bce}}(p, y) + \lambda_{topo}\mathcal{L}_{\text{evtopo}}(P_{\text{pred}}, P_{\text{true}}). \quad (9)$$

3.2.3 Gradient Computation of EvTopoLoss

Computing the Wasserstein distance between predicted and ground truth persistence points through one-to-one correspondence is both computationally expensive ($O(n^3)$ complexity) and non-differentiable, hindering gradient-based optimization. We observe that globally optimal matching is not necessary; local loss assignment suffices for effective training. Accordingly, we reformulate the topological loss by directly assigning gradients to each prediction:

$$\nabla_\pi \mathcal{L}_{\text{evtopo}} = \sum_{p \in \mathcal{C}(P_f)} \frac{\partial \mathcal{L}_{\text{evtopo}}}{\partial \pi(p)}, \quad (10)$$

where $\mathcal{C}(P_f)$ denotes the set of critical points in the filtered point cloud. However, directly computing gradients for all critical points is prohibitively expensive in large-scale point clouds. We propose an active region mechanism to optimize gradient computation, which is defined as:

$$\mathcal{A} = \{p_i \mid \tau_{\text{pred}} - \delta < \pi(p_i) < \tau_{\text{pred}} + \delta, p_i \in P\}, \quad (11)$$

where δ is the boundary width. Only points near the threshold are adjusted during training, thus we only compute gradients for these points. For points within the active region, the gradient is based on a few distance computations:

$$\frac{\partial \mathcal{L}_{\text{evtopo}}}{\partial \pi(p_i)} = \tanh(2 \cdot (d^P_{\min}(p_i) - d^T_{\min}(p_i))) \cdot \eta, \quad (12)$$

where $d^P_{\min}(p_i) = \min_{p_j \in P_f} \|p_i - p_j\|$ and $d^T_{\min}(p_i) = \min_{t_j \in T_f} \|p_i - t_j\|$ measure distances from point p_i to the ground truth point cloud. The gradient mechanism adaptively modulates point retention: active points with smaller distances to the ground truth receive positive gradient signals encouraging retention, while those proximal to the prediction yet distant from the ground truth are suppressed via negative gradients promoting exclusion. To further improve efficiency, we perform batch processing on the active region:

$$\mathcal{A} = \bigcup_{k=1}^{\lceil |\mathcal{A}|/B \rceil} \mathcal{A}_k, \quad |\mathcal{A}_k| \leq B. \quad (13)$$

Our method focuses gradient computation on boundary regions near the decision threshold through the active region mechanism, reducing the number of critical points that need processing. Meanwhile, geometric distance guidance directs gradients toward persistent features and their corresponding critical points, maintaining topological consistency with ground truth.

4. Experiments

4.1. Experimental Settings

Dataset. To the best of our knowledge, EV-UAV [4] is currently the only large-scale, highly diverse benchmark for event-based small object detection (EVSOD). The dataset contains 147 meticulously annotated sequences totaling over 20.3 million events, capturing diverse UAV maneuvers (forward/backward, lateral, vertical, and combined motions) across varied lighting conditions (normal, bright, low-light) and backgrounds (urban, mountainous, forested, aquatic, and aerial environments). Extremely small target scales (most smaller than 32×32 pixels) render this dataset particularly challenging for detection algorithms. We conduct experiments based on the official data splits.Following the original evaluation criteria [4], we assess segmentation performance using Intersection over Union (IoU) and accuracy (ACC). For detection performance, we assess the results via probability of detection (P_d) and false alarm rate (F_a), which are derived from the segmentation outputs.

Implementation Details. Our network is trained for 50 epochs with batch size 1 using the Adam optimizer [14]. The learning rate is initialized to 0.001 with a step decay of 0.1 every 10 epochs. For computing EvTopoLoss, τ_{pred} and τ_{true} are both set to 0.5 as the standard binarization thresholds [26]. Persistent homology features up to dimension 1 (H_0 and H_1) are extracted using Ripser [45]. We select the model with the best validation performance for final evaluation. All experiments are implemented in PyTorch on an NVIDIA RTX 3090 GPU.

4.2. Comparison with SOTA

Quantitative Results. The results in Tab. 1 indicate that SpTopoNet achieves superior detection performance compared to previous methods. We attribute this to three reasons. Firstly, the explicit modeling of local-global topological structures in the network architecture substantially enhances feature discrimination. Secondly, the topological regularization incorporated in the loss function effectively suppresses spurious detections. Finally, our active region gradient computation strategy ensures computational efficiency by concentrating topological loss calculation near decision boundaries, adding minimal overhead while delivering improvements.

Qualitative Results. The visualization results are presented in Fig. 5. Our method achieves the best performance in maintaining the continuity of small moving object trajectories. Especially in complex scenarios with dense background noise (as shown in the first rows), the trajectories detected by our method are more complete and continuous. Moreover, in cases of multi-target complex motion (as shown in the second to fourth rows), our method also shows clear improvements. We attribute this enhancement to the incorporation of topological constraints in both the local feature extraction module and the loss function, which enables the network to better preserve the topological structure of object trajectories.

4.3. Ablation Studies

4.3.1 Contribution of Components

As shown in Tab. 2, incorporating TLM into the baseline yields significant performance improvements, indicating the critical role of local spatiotemporal topological features. The comparable gains achieved by SCM alone demonstrate the importance of global contextual information. Their combination produces further improvements as local and global features complement each other. The incorporation of EvTopoLoss yields superior performance, which can be attributed to the preservation of structural integrity in target events through topological constraint.

Table 1. Quantitative comparison of the proposed method to state-of-the-art methods.

Methods	Publications	Event Rep.	IoU(%)↑	ACC(%)↑	P_d(%)↑	F_a(10^{-4})↓	#Params.	Runtime (ms)
SSD [23]	ECCV2016	Event Count	25.31	28.56	26.31	486.63	25.2M	2113
Faster RCNN [38]	TPAMI 2016	Event Count	26.93	29.68	27.39	689.68	41.2M	3962
DETR [56]	ICLR2020	Event Count	30.35	33.63	31.64	631.37	39.8M	3136
YOLOV10-S [47]	NIPS 2024	Event Count	32.55	33.39	32.18	589.67	7.3M	1627
EMS-YOLO [43]	ICCV2023	SNN	36.77	42.92	50.68	112.36	3.3M	1229
Spike-YOLO [25]	ECCV2024	SNN	43.94	48.26	59.62	55.38	69.0M	1883
GET [32]	ICCV2023	Group Token	40.31	48.91	60.73	46.35	18.4M	2168
RED [33]	NeurIPS 2020	Voxel Grid	35.99	45.54	53.76	102.27	24.1M	3427
RVT [10]	CVPR 2023	Voxel Grid	43.21	51.38	60.35	55.68	9.9M	1737
SAST [31]	CVPR 2024	Voxel Grid	34.31	40.22	51.21	150.32	18.5M	3075
KPConv [44]	ICCV2019	Points	48.19	57.28	68.59	16.32	50.1M	562
RandLA-Net [12]	CVPR2020	Points	50.32	59.29	70.56	6.95	**1.2M**	353
COSeg [1]	CVPR 2024	Points	51.89	60.93	71.32	9.21	23.4M	364
Ev-SpSegNet [4]	ICCV 2025	Points	<u>55.18</u>	<u>65.02</u>	<u>77.53</u>	<u>1.63</u>	<u>4.0M</u>	**35.9**
Ours	—	Points	**66.62**	**74.43**	**83.36**	**1.29**	4.4M	<u>56.5</u>

The **bold** and the <u>underline</u> represent the best and second-best performance, respectively.

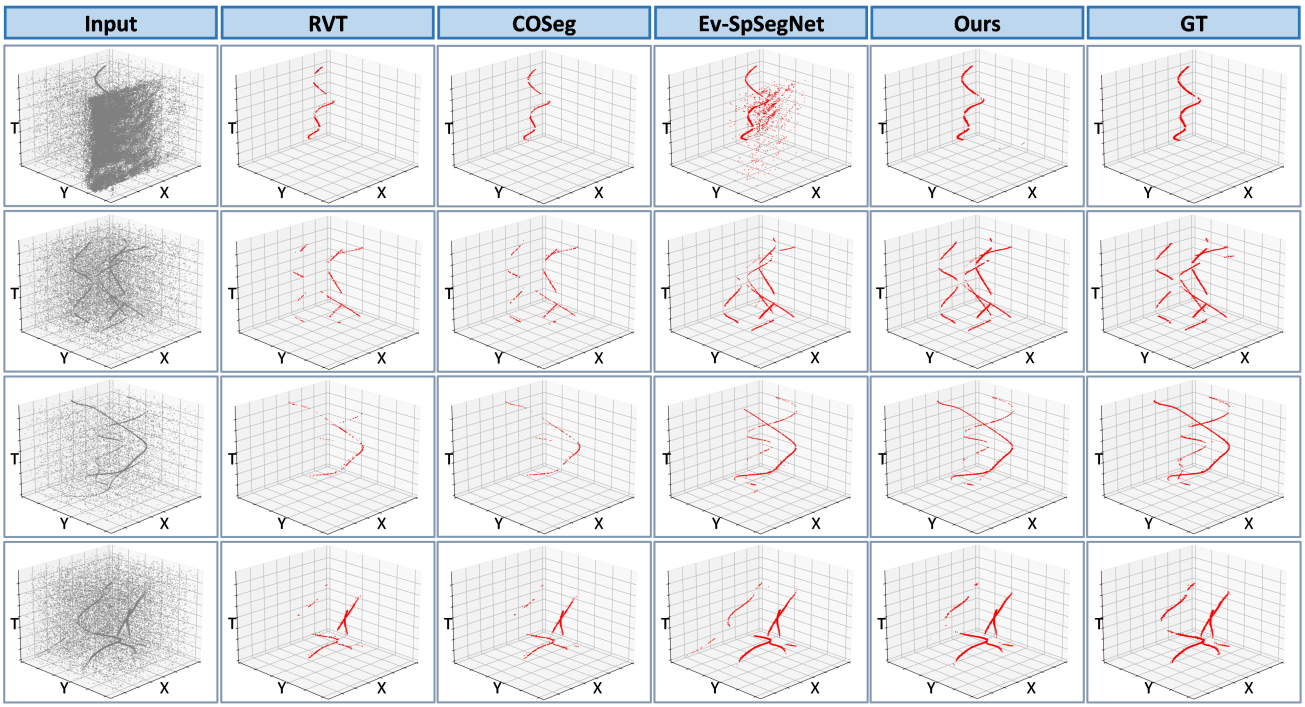


Figure 5. Qualitative comparison on the EV-UAV dataset. Our method achieves more continuous trajectories with fewer false alarms.

The visualization results in Fig. 6 show that the baseline suffers from severe trajectory fragmentation. In contrast, TLM and SCM leverage local and global topological structures to maintains trajectory continuity and suppress false alarms. The topological loss $\mathcal{L}_{\text{evtopo}}$ guides the network to preserve the structural consistency between predicted event point clouds and ground truth.

4.3.2 The Impact of TLM Layer Location

We conducted ablation experiments to analyze the effectiveness of applying TLM in different encoding layers, as shown in Tab. 3. TLM consistently improves performance across all layers, demonstrating the universality of topological learning on multiple feature scales. Shallow layers (particularly layer 1) yield the most substantial improvements.

Table 2. Ablation study on framework scheme and loss function.

TLM	SCM	$\mathcal{L}_{\text{evtopo}}$	IoU(%)↑	ACC(%)↑	P_d(%)↑	$F_a(10^{-4})$↓
			52.77	58.41	75.15	2.31
✓			63.32	69.50	81.01	1.73
	✓		62.50	68.98	81.68	1.49
		✓	61.47	66.55	77.23	1.44
✓	✓		64.07	69.73	79.66	1.51
✓	✓	✓	**66.62**	**74.44**	**83.37**	**1.29**

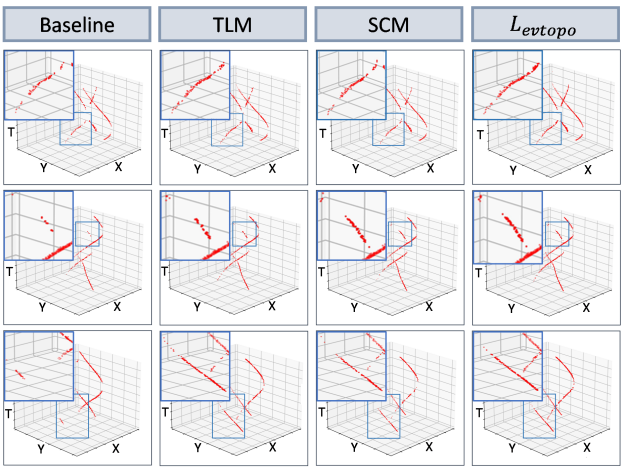


Figure 6. Trajectory visualization for ablation study. Compared with the baseline that produces fragmented results, each proposed component (TLM, SCM, $\mathcal{L}_{\text{evtopo}}$) independently improves trajectory smoothness and continuity.

Table 3. The influence of the Topological Learning Module.

TLM layers	IoU(%)↑	ACC(%)↑	P_d(%)↑	$F_a(10^{-4})$↓	#Params.
None	60.64	68.07	76.60	4.31	**4.1**M
layer 1	64.80	70.92	81.44	1.41	4.2M
layer 2	61.57	70.39	79.81	1.99	4.2M
layer 3	61.60	68.18	79.35	2.31	4.2M
layer 1,2,3	**66.62**	**74.44**	**83.37**	**1.29**	4.4M

This can be attributed to their preservation of richer spatial structures and neighborhood connectivity, which are essential for topological learning. Deeper layers tend to discard fine-grained geometric details during feature abstraction. The multi-layer combination (layers 1, 2, 3) achieves optimal performance. Shallow layers capture local topological structures while deeper layers encode semantic patterns, effectively balancing topological preservation with semantic understanding.

4.3.3 The Effectiveness of EvTopoLoss

To validate the generalizability of EvTopoLoss, we integrate it into three methods: Ev-SpSegNet [4], 3D-UNet [6], and PointNet++ [35]. As shown in Tab. 4, Ev-SpSegNet with $\mathcal{L}_{\text{evtopo}}$ improves IoU, Acc, and P_d by 8.49%, 3.78%, and 4.91%, respectively, and reduces F_a by 21.76%. Comparable improvements are achieved regardless of the backbone architecture employed. These results demonstrate that EvTopoLoss effectively enhances existing methods by maintaining global trajectory structure through topological relationship constraints in event point clouds, leading to more accurate segmentation and fewer false alarms.

Table 4. The performance of EvTopoLoss on different methods.

Methods	IoU(%)↑	ACC(%)↑	P_d(%)↑	$F_a(10^{-4})$↓
Ev-SpSegNet	53.62	64.02	76.76	1.93
Ev-SpSegNet*	**62.11**	**67.80**	**81.67**	**1.51**
3D-unet	48.32	56.38	64.29	4.86
3D-unet*	**53.22**	**61.58**	**67.43**	**3.80**
PointNet++	35.44	48.67	31.07	22
PointNet++*	**36.32**	**49.99**	**39.86**	**21**

* represent the methods with $\mathcal{L}_{\text{evtopo}}$.

Table 5. Ablation study on weight λ_{topo} on validation set.

λ_{topo}	IoU(%)↑	ACC(%)↑	P_d(%)↑	$F_a(10^{-4})$↓
0.01	58.33	66.47	70.32	4.36
0.03	63.95	71.61	73.95	2.20
0.05	**67.17**	**75.26**	**80.47**	**1.01**
0.1	54.19	67.64	72.37	2.91

4.3.4 Ablation Studies on Weight λ_{topo}

We conducted experiments on the validation set to determine the optimal weight λ_{topo} (Eq. (9)) for the topological loss function. As demonstrated in Tab. 5, insufficiently small values of λ_{topo} yield marginal improvements, whereas excessively large values tend to induce convergence difficulties and deterioration in model performance. Following this ablation study, we select $\lambda_{topo} = 0.05$ and apply it to all subsequent evaluations on the test set.

5. Conclusion

In this paper, we propose SpTopoNet, a topology-constrained sparse convolutional network for small moving object detection with event cameras. By modeling the topological structure of motion trajectories in event point clouds, our method ensures spatiotemporal continuity in object detection results. Furthermore, the proposed EvTopoLoss utilizes topological constraints to maintain trajectory structural integrity, which enables the network to retain more target events and produce more complete detections. In the future, we will continue to explore topological analysis in event point clouds, extending this framework to multi-object tracking and more complex dynamic scenarios.

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